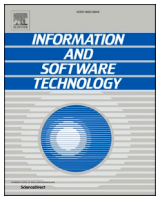
Information and Software Technology 198 (2026) 108209



Contents lists available at ScienceDirect

Information and Software Technology

journal homepage: www.elsevier.com/locate/infsof





Challenges to lean principles adoption in software project management: A study of software development firms in developing countries

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ARTICLE INFO

Keywords:
Lean principles
Software project management
Software development industry
PLS-SEM

STRUCTURED ABSTRACT

Context: Lean principles, widely recognized for improving efficiency, have gained global relevance and are often integrated into Agile software project management methodologies such as Kanban. However, their adoption in developing countries remains challenging due to resource constraints and organizational limitations. In particular, the software development industry in such contexts faces barriers that hinder effective Lean implementation.

Objectives: This study seeks to investigate the challenges of Lean adoption in software project management within Sri Lanka's software development sector. By focusing on the managerial perspective, it emphasizes the critical role project managers play in driving Lean practices. The research aims to identify key barriers to adoption and provide insights that contribute to both theory and practice in developing economies.

Methods: A conceptual framework was developed through an extensive literature review, grounded in three organizational theories: Resource-Based View, Contingency Theory, and Institutional Theory. Data was collected through a structured survey administered to professionals in Sri Lanka's software development industry. The survey, based on a five-point Likert scale, was designed to capture challenges to Lean adoption. Partial Least Squares Structural Equation Modeling (PLS-SEM) was employed to quantitatively analyze the data and test the framework.

Results: The results validate three major barriers to Lean adoption: Lean Specific Knowledge (LSK), Short-term Economic Benefits (SEB), and Lean Supportive Culture (LSC). Among these, LSK emerged as the most significant factor impeding Lean adoption, followed by SEB and LSC.

Conclusion: The study contributes to the literature by applying established organizational theories to the context of developing economies' software industries, thereby extending their applicability. Practically, it provides actionable insights for software project managers and organizational leaders seeking to overcome Lean adoption barriers. Enhancing Lean knowledge, fostering supportive cultural practices, and aligning economic perspectives with long-term benefits are essential for successful Lean integration.

1. Introduction

The lean principles originated from the automobile manufacturing industry, aiming at enhancing value by eliminating non-value-adding activities [1]. Over time lean philosophy has been adopted by various industries such as Manufacturing, Process Industries, Small and Medium Enterprises, Aerospace, Product development, Textile, Construction, Services, and Agri-business [2]. Software companies seek innovative approaches to enhance productivity and meet evolving customer demands for high-quality, cost-effective solutions. Customers expect quality code with innovative features at a low cost. Lean principles have been considered to meet these requirements. A case study found that lean techniques, which revolutionized manufacturing, can also transform software development by minimizing work in progress. These techniques offer feedback to improve performance by addressing internal service barriers and organizational structure issues in software development [3].

The adoption of Lean principles in software project management has recently become a significant trend. Agile methodologies, such as Kanban and Lean development, are rooted in Lean philosophy. The

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https://doi.org/10.1016/j.infsof.2026.108209
Received 29 August 2025; Received in revised form 5 April 2026; Accepted 27 May 2026
Available online 28 May 2026
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software development industry has demonstrated the application of Lean both as an independent methodology and as part of a hybrid approach, combining Lean principles with other software development methodologies [4]. Lean and Agile philosophies have distinct origins but share some overlapping ideas. Agile prioritizes flexibility and the ability to quickly adapt to change, while Lean emphasizes overall economic contribution from a holistic enterprise perspective. In software engineering, discussions about Lean thinking began in the 1990s, predating the Agile Manifesto, with Lean often cited as an inspiration for Agile principles [5,6].

The global software development industry is expanding rapidly, with market projections reaching approximately $741 billion by 2025 [7]. Mirroring this growth, Sri Lanka's software sector is experiencing notable momentum, with an anticipated annual growth rate of 5.26% from 2025 to 2030. The Sri Lankan ICT sector aims to contribute $15 billion to the digital economy by 2030. In 2022, it earned $1.5 billion in foreign exchange and employed around 175,000 skilled professionals, making it the country's fourth-largest export earner with over 300 active firms [8]. Given these positive trends, Sri Lanka represents a strong candidate in the context of developing countries to study Lean adoption in software project management. Its rapidly growing software development industry offers a relevant and timely setting to investigate practical barriers to Lean implementation, addressing both local and broader regional knowledge gaps.

In recent years, the software development industry, primarily influenced by the agile community, has begun to explore Lean methodologies, with many firms adopting agile and/or Lean practices, positioning Lean as a relatively new entrant in this context [9]. Despite this growing interest, the universal applicability of core Lean principles in creative and knowledge-intensive domains such as software development remains contested [10]. Furthermore, [11] argue that the successful implementation of Lean in the IT sector is inherently difficult and emphasize the scarcity of IT industry-specific empirical studies. This limitation is particularly pronounced in developing countries, where [12] note that the adoption of Lean faces distinct challenges that require contextual investigation. Consequently, there is a need to examine the practical barriers to Lean implementation within the software development industry, especially in the context of developing economies, to address this underexplored area in the existing body of knowledge.

To address this identified research gap, the present study aims to investigate the common challenges associated with adopting Lean principles in the software development industry of developing countries, using a multi-theoretical lens. By integrating insights from several organizational theories, the study seeks to enhance the analytical depth in understanding these barriers. Specifically, the study pursues the following research objectives: (RO1) To develop a theoretical framework to identify the main types of challenges in adopting Lean principles for software project management; (RO1.1) To identify the specific challenging factors related to Lean adoption; and (RO2) To systematically examine the impact of these identified challenges on Lean adoption within the software development industry. Drawing on a comprehensive literature review and structured quantitative analysis, the research validates and prioritizes the most critical obstacles hindering Lean adoption. Through this systematic investigation, the study contributes to a more nuanced understanding of the practical impediments to Lean implementation, offering valuable insights for both academic research and industry practice within the software sector of developing economies. The remainder of this paper is organized as follows. Section 2 presents the literature review and establishes the theoretical foundation of the study. Section 3 describes the methodology, including the conceptual framework, questionnaire development, data collection, and analytical approach. Section 4 presents the data analysis and empirical results. Section 5 discusses the main findings in relation to prior literature and outlines the practical and theoretical implications of the study, followed by limitations and directions for future research.

2. Literature review

2.1. Lean principles

The concept of "Lean" originated from the Massachusetts Institute of Technology's International Motor Vehicle Program, designed to describe the waste-reduction methods of the Toyota Production System, in contrast to traditional craft and mass production methods [1]. The National Institute of Standards and Technology defines Lean as a systematic approach focused on eliminating waste through continuous improvement, enabling smooth product flow based on customer demand, and striving for perfection [13]. further expands on Lean philosophy by identifying eight common types of waste (non-value-adding activities), including: overproduction, waiting, transportation, non-value-added processing, excess inventory, defects, excess motion, and underutilized people. To address these wastes, several essential Lean tools are recommended: Pull System for demand-driven production, Kanban for efficient material flow, Work Cells for optimal workforce placement, Total Productive Maintenance (TPM) for equipment efficiency, Total Quality Management (TQM) for continuous process improvement, Workplace Organization (5S), Visual Controls for immediate status awareness, and Concurrent Engineering to facilitate cross-functional collaboration in product or service development.

Lean principles require an understanding of technical aspects and dependencies between various functional areas: Engineering, Human Resources, and Marketing within an organization [14]. Ohno, one of the founders of the Toyota Production System [15], identifies two core principles of lean production: Just-In-Time (JIT) and autonomation. JIT is a "pull" system where production at each stage is initiated only in response to a downstream demand signal, facilitated by the "Kanban" method governed by six specific rules. Autonomation, also known as "Jidoka", integrates automation with human oversight, aiming to concentrate production workers on the most value-added activities rather than eliminating their roles. Following Ohno's findings, [16] present rules capturing four essential principles of lean systems: Specific tasks, Streamlined communication, Simple process architecture, and Hypothesis-driven problem-solving.

Lean principles have been adapted to suit the unique characteristics of the software development industry [17]. outlines four fundamental principles tailored to this domain: eliminating waste by identifying and avoiding non-value-adding activities, focusing on people who add value through skill development and training, implementing a demand-pull system where development is driven by customer needs, and ensuring organization-wide adoption of lean practices. In the software context, seven key forms of waste are identified: overproduction (unnecessary features), inventory (backlogged requirements), additional processing (excessive documentation and approval), unnecessary motion (searching for information), defects (quality issues undetected by both QA and developers), waiting time (delayed feature delivery), and transportation (handoffs between teams and customers). Complementing this, [18] found that in U.S.-based SMEs, Lean is adopted primarily to reduce costs, minimize waste, promote continuous improvement, and optimize operations. Key drivers for lean implementation include: cost efficiency, better resource utilization, enhanced responsiveness, and sustained competitive advantage. Common lean tools applied in practice include: workplace organization, quality certifications, standardized workflows, and visual controls. The reported benefits of Lean adoption include: increased productivity, improved customer satisfaction, and reduced operational costs.

2.1.1. Lean adoption in Sri Lanka

In the Sri Lankan context, several studies have explored the adoption of Lean principles across different industries [19]. conducted a quantitative study within the export apparel manufacturing sector, identifying seven commonly implemented Lean practices that contribute to enhanced performance: waste minimization, continuous improvement,

defect minimization, Just-In-Time strategies, cross-functional teams, employee involvement, and effective information availability. The study emphasizes that with sustained focus on continuous improvement, these Lean practices are expected to mature further, leading to long-term performance gains within the industry.

Other sector-specific studies also highlight Lean implementation in Sri Lanka [20], through a case study of small and medium-sized enterprises, identified low-resource-intensive Lean tools such as: 5S, root cause analysis, value stream mapping, work standardization, statistical process control, and quality circles. In contrast, more complex tools like Kanban, small batch sizing, and pull systems were noted as resource-intensive. The study revealed several implementation barriers in SMEs, including limited financial resources, insufficient Lean knowledge, and a lack of institutional support. Similarly, [21] applied the Delphi Technique to assess Lean practices in the Sri Lankan construction industry, identifying common waste types and confirming the relevance of Lean principles such as waste elimination, customer value focus, and a culture of continuous improvement. Both studies were limited to qualitative analysis.

2.1.2. Lean services vs manufacturing

Lean practices differ notably between the service and manufacturing sectors due to the inherent characteristics of each. Alsmadi et al. [22], in a UK-based comparative study, examined ten Lean practice constructs related to internal operations, supply, and customer focus. Service firms outperformed manufacturing firms in employee and customer involvement, highlighting the service sector's reliance on human engagement to enhance service quality. However, practices such as TPM, supplier feedback and development, and setup time reduction were found to be irrelevant to service sector performance. The study identified six Lean practices: customer involvement, JIT delivery by suppliers, pull systems, continuous flow, Statistical Process Control (SPC), and employee involvement, as significantly contributing to service sector performance. Despite contextual differences, both sectors demonstrated performance improvements through Lean adoption.

Supporting this, [23] conducted a literature review identifying common Lean tools across both sectors, such as TQM, JIT, Kaizen, Six Sigma, SPC, TPM, and Kanban. Tools like work teams and job reengineering were more prominent in the service sector, reflecting its focus on human-centered processes. Further, [24] adapted Lean manufacturing principles for services by redefining waste and value based on the unique characteristics of services. Key Lean Service principles included customer-defined value, optimizing service flow, and striving for perfection. The study also emphasized continuous customer engagement and identified service-specific wastes, such as overproduction due to poor planning, duplication of effort, and lack of standardization. These findings confirm that while Lean can be adapted to service contexts, its implementation must be tailored to address service-specific challenges and objectives.

2.1.3. Adopting lean principles - related barrier studies

A range of international studies highlights significant barriers to Lean implementation, particularly in technology and manufacturing sectors. In the Indian IT industry, [11] identified eleven barriers to Lean Six Sigma adoption through a quantitative study, with the most critical being the lack of continuous involvement in Lean projects and insufficient time for implementation. Additional challenges included high staff turnover, data management difficulties, vague project scopes, extended timelines, and unclear process parameters. Interestingly, although knowledge and training-related barriers were considered less impactful overall, they were highly significant for less experienced technical staff. The study recommends integrating Lean Six Sigma tools into the core software development process, but was limited to data from a single organization. Similarly, [25], through research conducted in Greece, identified financial constraints and workforce-related issues such as lack of leadership commitment and organizational resistance to change as primary barriers. They concluded that Lean must be embedded within the organizational culture to be successful.

Further insights are provided by [26] in Brazil's manufacturing sector, where the lack of management support, insufficient accountability among operators, and the absence of skilled Lean practitioners were noted as critical obstacles. Their qualitative case study recommended a structured implementation framework to address these challenges. Recently, [12] conducted a quantitative study in Malaysia's furniture industry using PLS-SEM to validate three main barrier types: knowledge-related, cultural/attitudinal, and resource-related. Among these, knowledge-related barriers such as lack of Lean expertise and practical understanding were found to have the most significant impact. Cultural resistance and high investment requirements also hindered Lean adoption. The study emphasizes the need for broader Lean awareness through educational and governmental initiatives. Collectively, these findings reveal the multifaceted and context-specific nature of Lean implementation challenges.

2.2. Software project management

Software project management is conceptualized as the process of aligning software solution attributes such as requirement certainty, risk nature, complexity, team competency, and technological understanding, with the most appropriate development life cycle and management approach to maximize business value [27]. Through a systematic literature review, [28] identified seven critical practices for effective global software project management, including role clarity, use of project management tools, agile practices, and trust-based team dynamics [29]. introduced the "4PTRB" framework, encompassing People, Processes, Products, Projects, Technology, Risk, and Business management areas. Their research underscores that effective project management is a key success factor in navigating resource constraints, strict timelines, and evolving technologies. The PMBOK Guide and CMMI-DEV view project management as a set of processes; accordingly, past literature derives software project management frameworks such as 3P's, 4P's, and 3PR, encompassing areas such as People, Process, Product, Project, Technology, and Risk [30].

2.2.1. Software project management methodologies

Software project management methods vary in structure and suitability based on project requirements and complexity. Bogdan-Alexandru ANDREI et al. [31] reflects that the traditional Waterfall model, originating from manufacturing industries, divides the project into sequential phases and is effective only when requirements remain stable throughout the development process. However, due to the dynamic nature of software projects, Agile methodologies characterized by flexibility, continuous feedback, and iterative development are more widely adopted. Kanban focuses on visualizing workflow and managing work-in-progress, while Scrum emphasizes iterative task prioritization and team collaboration based on values like commitment and openness. Agile's adaptability makes it preferable for evolving projects. Al-Saqqa et al. [32] explains Agile as a philosophy encompassing various methods such as Scrum, Test-Driven Development, Feature-Driven Development, Extreme Programming, and Dynamic Systems Development Method. Agile promotes incremental development, close customer collaboration, and minimal overhead, especially benefiting small to mid-sized projects. Despite its advantages, there is no one-size-fits-all approach; the chosen methodology must align with the project's size, goals, and changing constraints to ensure effectiveness.

2.2.2. Lean software project management

Lean software project management emphasizes efficient, value-driven processes through structured principles and continuous improvement. Middleton et al. [33] identify key Lean principles in software projects, including maintaining low work-in-progress, applying Just-In-Time practices, workload balancing, visual controls, and using

automation tools for rapid delivery. Unlike Agile's people-focused approach, Lean is data-driven, employing statistical process control to improve predictability. Lean also promotes multiskilling and collaboration, allowing team members to address bottlenecks beyond their core expertise. These practices reduce lead time, variability, and error rates while enabling certification readiness and higher profitability. Poppendieck et al. [34] reinforce Lean's conceptual grounding with seven core principles: customer-defined value, waste elimination, built-in quality, continuous learning, rapid incremental delivery, employee involvement, and ongoing improvement. Lean integrates the full product lifecycle from design to validation into a cohesive feedback loop, fostering sustained value delivery.

2.3. Lean applied to the software development industry

The application of Lean principles in the software development industry differs significantly from traditional lean manufacturing practices. Yadav et al. [2] highlight that while lean manufacturing involves visible wastes, tangible deliverables, and predictable workflows, lean software development deals with intangible inventories, invisible waste, vague requirements, and variable task structures. Of the five foundational lean principles, Value, Value-stream, Flow, Pull, and Perfection, only Value and Flow were found highly applicable to software development. Pernstål et al. [35] extend this perspective through the lens of Lean Product Development (LPD), which adopts an enterprise-level application of lean principles. The study emphasizes the importance of reducing waste through value stream mapping, promoting customer-defined value, and using visual management. However, principles like supplier integration, standardized tools, and continuous improvement were found to be less impactful. Common wastes in large-scale software development include excessive code backlogs, defects, and non-value-adding features.

[9] quantitatively analyze lean adoption in the Finnish software industry and discover that, although Agile practices are more common, Lean principles such as eliminating waste, ensuring quality, and empowering people are also widely adopted. The main motivations include productivity, risk reduction, and improved development flow, while challenges include a lack of management support, cultural resistance, and skill shortages. Similarly, [4] examine how Lean complements Agile in software firms. They propose five integration strategies, with "Lean within Agile" being the most adopted. Key Lean practices include Kanban, value stream mapping, waste elimination, bottleneck management, and customer-defined quality. The study concludes that Lean enables strategic-level improvements, enhancing Agile methods and driving enterprise-wide efficiency.

2.3.1. Knowledge transfer and cultural change in software development firms

Knowledge transfer and cultural change are central to enabling sustained process improvement in software development and IT service firms. Evidence from software process improvement research indicates that an organization's internal culture and leadership behavior shape whether knowledge is shared and translated into improved routines [36]. In the context of Lean adoption, this implies that improvement practices are more likely to be sustained when leadership reinforces learning behaviors, allocates time and resources for improvement work, and normalizes knowledge sharing, allowing new routines to spread beyond individual champions [36].

Software process knowledge management research further emphasizes that improvement depends on deliberately managing process knowledge. Meehan and Richardson describe software process knowledge management as focusing on identifying and managing process-related knowledge that supports improvement initiatives [37]. When interpreted through a Lean adoption lens, this indicates that firms require a stable foundation of Lean-related knowledge and that this knowledge should be embedded into day-to-day practices rather than remaining tacit within a small subgroup [37].

In agile software organizations, knowledge sharing is often strong within teams but weaker across teams, limiting the diffusion of organization-wide process change [38]. A survey study in a large agile organization reports that cross-organizational knowledge sharing frequently requires explicit mechanisms and incentives [38]. For Lean adoption, this suggests that cultural conditions supporting collaboration across teams and shared improvement goals are necessary to scale Lean practices beyond local teams [38].

Global and distributed software development studies similarly indicate that knowledge transfer is constrained by distance and coordination barriers. Practitioners often adopt targeted techniques and strategies to enable sharing across sites [39]. Site visits have also been shown to play a knowledge-based role in reducing barriers and strengthening shared understanding in global software development settings [40]. For Lean adoption, these findings imply that cultural change should address coordination norms and trust-building practices that enable knowledge transfer across boundaries, ensuring that improvement-related learning does not remain fragmented across locations or teams [39,40].

Across software development and IT service contexts, prior work consistently treats knowledge-related capability, organizational culture, and Lean-implementation outcomes as conceptually separable dimensions of process change. Studies on Lean and Agile use in software firms distinguish capability constraints, such as skill shortages and limited methodological expertise, from cultural constraints such as weak managerial commitment and resistance to change, while assessing adoption through the extent to which Lean-oriented practices become routine [9,11]. Software process improvement research similarly positions knowledge sharing as an enabler that depends on leadership support and cultural norms and evaluates success through downstream improvements in process performance rather than through the antecedent conditions themselves [36]. Evidence from large agile organizations further indicates that knowledge exchange can be strong within teams yet weak across teams, implying that collaboration norms and coordination routines are distinct from individual or team-level know-how and from the realized outcomes of process change [38]. In distributed software development, practices such as site visits are discussed as mechanisms for transferring knowledge and building shared understanding across boundaries, again separating enabling conditions from implementation results [40]. Overall, this body of evidence supports analyzing knowledge resources, cultural/organizational support for change, and realized lean-adoption outcomes as related but empirically distinguishable constructs in software settings. Table 1 summarizes the principal studies reviewed in this section and highlights their main findings relevant to lean adoption and software project management.

3. Methodology

This study employs a positivist research philosophy, a correlational research design, and deductive reasoning to investigate the challenges associated with Lean adoption in software development firms. A conceptual framework, derived from a comprehensive literature review, is empirically tested using PLS-SEM, a technique consistent with similar lean adoption studies [12]. Data were gathered through structured questionnaires administered to employees within software development organizations, facilitating a quantitative evaluation of critical barriers.

3.1. Hypotheses and conceptual framework development

3.1.1. Conceptual framework

A conceptual framework (Fig. 1) was developed to examine the challenging factors associated with adopting Lean principles in software project management. The framework categorizes three antecedent challenging factors under three organizational theories: Resource-Based View (F1), Contingency Theory (F2), and Institutional Theory (F3), operationalized as Lean-Specific Knowledge (LSK), Short-term

Table 1
Key literature summary.

Study	Context	Key findings
[13]	General lean principles	Identified 8 common types of waste and core lean tools such as pull systems, Kanban, TPM, TQM, 5S, and visual controls.
[15]	Toyota Production System	Identified 2 core principles of lean production: Just-In-Time and autonomation.
[16]	Lean systems	Presented 4 essential principles: specific tasks, streamlined communication, simple process architecture, and hypothesis-driven problem-solving.
[17]	Software development	Outlined 4 lean principles for software and identified 7 software-specific wastes.
[19]	Sri Lankan apparel industry	Identified 7 lean practices contributing to performance improvement.
[20]	Sri Lankan SMEs	Identified low-resource lean tools such as 5S, root cause analysis, value stream mapping, work standardization, SPC, and quality circles; resource-intensive tools included Kanban, pull systems, and small batch sizing.
[22]	UK manufacturing and service sectors	Examined 10 lean practice constructs and found 6 practices significantly contributing to service-sector performance.
[11]	Indian IT industry	Identified 11 barriers to Lean Six Sigma implementation, with lack of continuous involvement and insufficient implementation time as the most critical.
[12]	Malaysian furniture industry	Validated 3 barrier categories: knowledge-related, cultural/attitudinal, and resource-related barriers; knowledge-related barriers were the most significant.
[28]	Global software project management	Identified 7 critical practices for effective software project management.
[29]	Software project management	Introduced the 4PTRB framework, covering people, processes, products, projects, technology, risk, and business management.
[34]	Lean software development	Reinforced 7 core principles of lean software development, including value, waste elimination, quality, learning, rapid delivery, employee involvement, and continuous improvement.
[2]	Software development	Reported that, among the 5 foundational lean principles, only Value and Flow were highly applicable to software development.
[9]	Finnish software industry	Found that Agile practices were more common, but lean principles were also widely adopted; key challenges included management support, cultural resistance, and skill shortages.
[4]	Lean and Agile in software firms	Proposed 5 integration strategies, with Lean within Agile being the most adopted.
[38]	Large agile organization	Reported that knowledge sharing was stronger within teams than across teams, indicating cross-organizational transfer difficulties
[36,37, 39, 40]	Software process improvement, knowledge management, and distributed development	Consistently emphasized the importance of leadership support, knowledge-sharing mechanisms, and coordination practices for sustaining process improvement and organizational change.

Economic Benefits (SEB), and Lean-Supportive Culture (LSC), respectively. These independent variables represent enabling conditions that influence Lean Principle Adaptation (LPA), which is conceptualized as a perception-based implementation outcome construct capturing outcome-oriented changes observable to employees. Accordingly, LPA is reflected through perceived implementation outcomes, including perceived reduction of Work-In-Progress (WIP) items, perceived adaptation of Just-In-Time (JIT) principles, and perceived continuous improvement through employee empowerment [9,35,33,34].

Lean-Specific Knowledge (LSK) refers to the extent to which personnel perceive that the organization possesses Lean-related know-how, training exposure, and implementation awareness that enable Lean adoption. Short-term Economic Benefits (SEB) refers to the extent to which personnel perceive that Lean adoption yields near-term financial and operational payoffs that justify continued effort and investment. Lean-Supportive Culture (LSC) refers to the extent to which personnel perceive that organizational norms, leadership support, and institutionalized routines encourage Lean-aligned behaviors. In contrast, the dependent construct, Lean Principle Adaptation (LPA), represents the perceived level of successful Lean adoption outcomes observable within the organization, and it captures perceived implementation outcomes rather than the enabling conditions. Accordingly, LSK, SEB, and LSC are conceptualized as antecedent challenging factors that influence LPA, while LPA reflects the perceived adoption outcome of Lean principles.

Table 2 outlines how the indicators associated with each independent variable align with the corresponding theoretical lens.

The Resource-Based View (RBV) emphasizes the importance of unique, valuable, and inimitable resources within an organization. Lean-related knowledge fits within this theory as it constitutes a critical organizational resource. These knowledge assets enable firms to improve processes, innovate, and maintain a competitive edge, thus justifying their relationship to RBV [41]. Drawing on this theoretical lens and supporting evidence from the literature, **F1** "Lean Specific Knowledge" (LSK) is defined as a critical factor influencing Lean adoption. This factor covers: lack of Lean-related knowledge and awareness; absence of formal guidelines; inadequate training and steep learning curve; need for specialized and technical skills; poor understanding of Lean tools, philosophy, and benefits [9,11,12,25,26]. Accordingly, it's hypothesized that Lean-Specific Knowledge positively affects the successful adoption of Lean Principles **(H1)**.

Contingency Theory posits that organizational effectiveness is achieved by fitting strategies to specific situational contingencies. The short-term economic benefits of Lean adoption enable organizations to adapt Lean strategies to their economic context, demonstrating the situational flexibility that Contingency Theory advocates [45]. Drawing on this perspective and the barriers identified in prior research, one primary factor affecting Lean adoption is proposed. **F2**, "Short-term Economic Benefits" (SEB) encompasses: Challenges in measuring Agile/Lean success and defining business value; high start-up and investment costs; unclear or delayed financial benefits; difficulty in quantifying financial gains; lack of perceived financial advantage [9,11, 12,25,26]. Based on these insights, Greater Short-term Economic Benefits is hypothesized to positively affect the successful adoption of Lean Principles **(H2)**.

Institutional Theory focuses on the influence of organizational norms, values, and culture on behavior and practices. A Lean-supportive culture is deeply rooted in the norms and values that guide organizational behavior. This relationship demonstrates how institutional norms and cultural acceptance are crucial for the successful adoption of Lean principles [48]. Drawing on this theoretical perspective and corroborating barriers identified in the literature, **F3**, "Lean Supportive Culture" (LSC), is recognized as a critical factor influencing Lean adoption. This factor encompasses: Resistance to change; traditional organizational culture; lack of management support or commitment; cultural and environmental barriers; negative perception of Lean; insufficient top and middle management support; lack of sustained motivation; poor

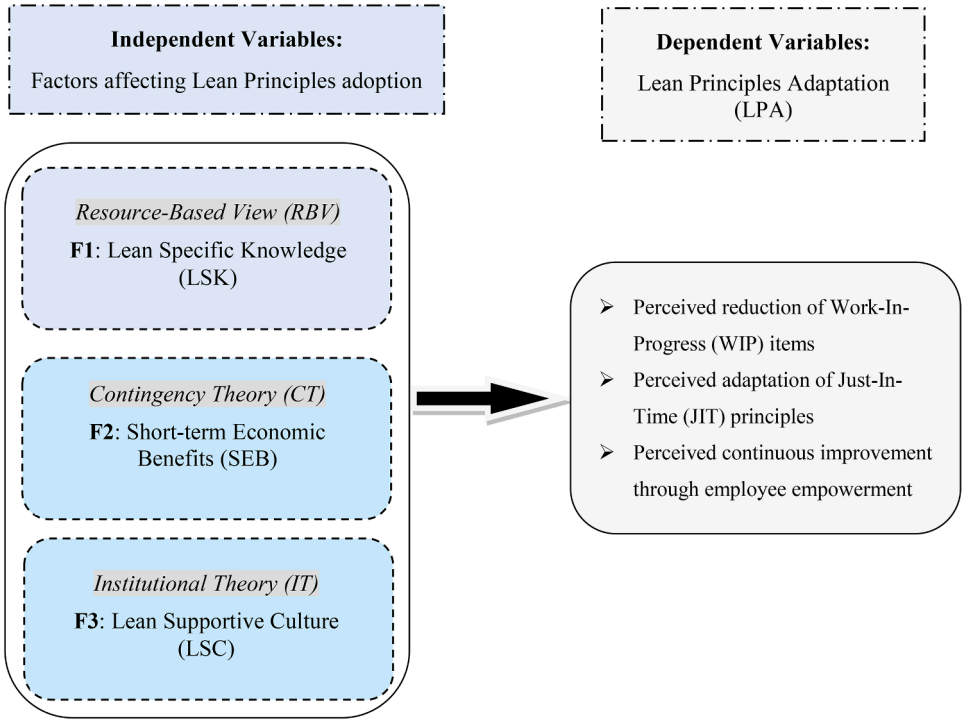


Fig. 1. Conceptual framework.

workforce engagement; absence of shared vision and urgency; skepticism about Lean benefits [9,11,12,25,26]. Accordingly, it's hypothesized that Lean Supportive Culture positively affects the successful adoption of Lean Principles (**H3**).

All constructs were specified using reflective indicators. With reflective measurement, the direction of influence is from the construct to the indicators, implying that the latent construct explains the covariation among the observed measures [51]. This specification is suitable when indicators are expected to be correlated and conceptually interchangeable, and when omitting a single indicator does not materially change the meaning of the construct [51]. Since the indicators in this study were developed as manifestations of underlying latent constructs and were intended to reflect the same conceptual domain within each factor, reflective measurement was retained for LSK, SEB, LSC, and LPA. In contrast, formative measurement assumes that indicators jointly form the construct and are not interchangeable, as each captures a distinct facet of the construct's domain [51]. This requires comprehensive domain coverage, and omitting an indicator may change the nature of the construct. In addition, collinearity among formative indicators can lead to unstable weights and nonsignificant estimates in PLS-SEM [51]. Given the intended interchangeability and expected intercorrelations among indicators in the present study, a formative model was not adopted.

3.2. Questionnaire development

The study investigates the impact of specific challenges on Lean adoption in software project management through a structured survey questionnaire. Designed using a five-point Likert scale, the questionnaire targets professionals in the Sri Lankan software development industry. It consists of 16 questions, four for each of the four key variables, using reflective indicators to ensure alignment with the conceptual framework. Table 3 presents the specific indicators, abbreviations used to generate the questionnaire, and the corresponding sources. These sources were selected after a thorough review of past literature on barrier studies related to the adoption of Lean principles. For transparency, the full questionnaire is provided in the Supplementary Information files. Before conducting the formal pilot test, which involved three experienced software project managers, the questionnaire underwent multiple rounds of feedback from two academic professionals (Senior lecturer, and Professor) and three software development industry professionals (QA Lead, Tech Lead, and Project Manager). This iterative process contributed to the refinement of the instrument by improving the precision and readability of the questions. A pilot test involving three experienced software project managers further ensured the instrument's effectiveness and eliminated any remaining ambiguities.

3.3. Data collection

This study adopts a quantitative, deductive research approach, consistent with methodologies used in similar Lean adoption barrier studies [12]. Hypotheses are developed and tested using survey data collected from professionals in the Sri Lankan software development industry. Based on recommendations by [52] a minimum of 37 responses is required to achieve 80% statistical power for detecting an R^2 value of 0.25 with a 5% error margin. Specifically, 58 invitations were distributed through industry contacts across >13 software development firms using Google Forms links circulated via professional networks and WhatsApp, resulting in a 76% response rate. Ultimately, 44 valid responses were obtained through the data collection. Partial responses were not recorded since all survey items were marked as mandatory before submission. Participants were recruited from employees holding lead-level positions or above, each with at least three years of experience in Sri Lankan medium to large-scale software development firms. This criterion was applied to ensure that respondents had adequate industry exposure and sufficient familiarity with Lean principles. Further, Table 4 summarizes the key demographic characteristics of the study participants, including gender, age group, marital status, highest education level, job title, years of experience, and company size. Due to resource limitations, convenience sampling is used, allowing efficient access to relevant participants. While this method may introduce bias and limit generalizability, it remains effective for exploring insights within the Sri Lankan context.

Table 2
Theoretical link of indicators.

Organizational Theory	Measurement Indicators	Theoretical Link
Resource-Based View (RBV) *Sources:* [41–44]	Familiar with Lean concepts	RBV explains competitive advantage through resources that are valued by the firm and are not readily obtainable by competitors. Familiarity with Lean concepts represents an initial knowledge resource that supports the development and preservation of firm-specific competences that can later influence visible performance dimensions such as cost, schedule, and quality in project delivery.
	Possess the know-how and experience of Lean implementation	RBV emphasizes that competitively relevant resources are those that remain difficult to imitate, substitute, or replicate. Lean implementation know-how and accumulated experience represent project management capabilities developed over time within a specific organizational environment, and such embedded capabilities are therefore positioned as strategic resources under RBV.
	Possess specialized knowledge and skills on Lean tools and their applications	RBV attributes inter-firm performance differences to heterogeneity in internal resources and capabilities. Specialized Lean tool knowledge and application skills constitute internal technical and human capabilities that can exert distinct effects on operational performance, supporting RBV's logic that unique practice bundles contribute to competitive advantage.
	Comprehends the benefits of Lean adoption	RBV highlights that resources contribute to strategic benefit when their value is recognized and prioritized for preservation, particularly when competitors would attempt to obtain similar competences. Comprehending Lean benefits reflects the firm's ability to attribute value to Lean-related knowledge resources, thereby supporting deliberate investment and retention of those competences within project management practice.
Contingency Theory (CT) *Sources:* [44–47]	Lean adoption yields definite cost reductions in the short-term	CT argues that project practices are adopted when they demonstrate congruence with dominant contextual demands. In software project environments characterized by tight delivery schedules and budgetary pressure, the ability of Lean practices to generate observable short-term cost reductions enhances their perceived fit with economic contingencies, thereby influencing managerial adoption decisions.
	Can obtain high profits and returns in short-term, through investments in Lean activities	From a CT perspective, managerial strategies are shaped by the need to respond rationally to environmental and financial conditions. Short-term profitability expectations reflect the alignment between Lean investments and immediate economic contingencies faced by software firms, where delayed or uncertain returns may reduce strategic attractiveness despite potential long-term benefits
	Short-term cost savings of Lean adoption can easily be quantified	CT emphasizes adaptation based on evaluative feedback mechanisms. In project-based software organizations, the quantifiability of short-term financial outcomes reduces uncertainty and enables managers to assess strategic fit more effectively. Easily measurable cost savings strengthen the perceived congruence between Lean practices and contextual decision-making requirements
	Requires only low initial investment cost to initiate Lean practices	According to CT, organizations must adjust their structural and strategic choices to match available resources and environmental constraints. In contexts where financial flexibility is limited, particularly in developing-country software firms, Lean practices with low initial investment requirements are more likely to be perceived as contextually appropriate and strategically feasible
Institutional Theory (IT) Sources: [44, 48–50]	Continuous support is provided by Top and Middle management to apply lean practices	IT explains adoption through regulative and normative elements that stabilize organizational practices via rules, surveillance, and obligation-like expectations. Continuous support from top and middle management functions as an internal institutional mechanism that legitimizes Lean as an accepted way of working within projects, shaping norms and reinforcing compliance and sustained participation. Such managerial endorsement also reflects normative pressures linked to professionalization and leadership expectations that influence whether Lean practices become institutionalized rather than remaining temporary initiatives.
	Perceiving lean implementation as easy and rapid	IT emphasizes cultural-cognitive elements such as shared beliefs and taken-for-granted assumptions that provide meaning and stability to organizational behavior.

(*continued on next page*)

Table 2 (*continued*)

Organizational Theory	Measurement Indicators	Theoretical Link
		When Lean is collectively perceived as easy and rapid to implement, it indicates cognitive legitimacy and cultural alignment, which facilitates institutionalization of Lean practices within project routines over time. In new contexts, adoption decisions can be bounded by social judgment and habitual interpretations rather than purely economic rationality, making shared ease perceptions central to whether Lean is accepted as "normal practice".
	Accepts the changes resulting from Lean adaptation	IT views organizational culture as socially constructed and continually translated through interactions among actors, where change acceptance depends on normative and cultural-cognitive alignment rather than technical feasibility alone. Acceptance of change therefore reflects the degree to which Lean-related behaviors are legitimized and compatible with the project's historically sedimented assumptions and shared meanings. In Lean implementation, socio-cultural forces may hinder full adoption even when economic incentives exist, indicating that cultural acceptance becomes a decisive institutional condition shaping Lean adoption outcomes
	Organizational-wide collaboration in adhering to lean principles	IT explains how collective adherence emerges through institutional norms and isomorphic pressures that encourage organizations to develop similar structures and practices. Organization-wide collaboration indicates that Lean is supported through shared norms and coordination mechanisms that extend beyond isolated teams, enabling Lean practices to become embedded across the organization. In practice, the adoption of Lean is shaped by the interplay of regulative, socio-cultural, and economic institutional forces, and collaboration becomes critical because institutional forces can be opposed, requiring coordinated effort to sustain Lean in the absence of uniformly supportive conditions.

3.4. Quantitative analysis

The data collected in this study is analyzed using Partial Least Squares Structural Equation Modeling (PLS-SEM), a technique widely applied in social science research for its suitability in handling complex models. PLS-SEM is selected over traditional tools such as SPSS due to several methodological advantages. This study involves constructs that cannot be directly measured, making PLS-SEM appropriate for testing the developed conceptual framework, which integrates multiple theoretical lenses. Unlike first-generation statistical tools that assume data normality, PLS-SEM is a nonparametric method that does not rely on distributional assumptions. This flexibility is particularly relevant as the variables under investigation may not follow a normal distribution. Additionally, the framework incorporates three organizational theories, and PLS-SEM is suited for evaluating their combined impact. It is also effective in achieving strong statistical power with smaller sample sizes, which is essential given the constraints of this study [51]. Given these strengths, PLS-SEM is adopted to conduct a comprehensive and robust analysis of the relationships among identified variables. This technique enables the exploration of complex theoretical constructs relevant to understanding the challenges of Lean adoption in software project management within the Sri Lankan software development context.

Table 3
Indicators and abbreviations.

Abbreviation	Description
LSK	**Lean Specific Knowledge** **Sources:** [9,11,12,25,26]
LSK1	Familiar with Lean concepts
LSK2	Possess the know-how and experience of Lean implementation
LSK3	Possess specialized knowledge and skills on Lean tools and their applications
LSK4	Comprehends the benefits of Lean adoption
SEB	**Short-term Economic Benefits** **Sources:** [9,11,12,25,26]
SEB1	Lean adoption yields definite cost reductions in the short-term
SEB2	Can obtain high profits and returns in short-term, through investments in Lean activities
SEB3	Short-term cost savings of Lean adoption can easily be quantified
SEB4	Requires only low initial investment cost to initiate Lean practices
LSC	**Lean Supportive Culture** [9,11,12,25,26]
LSC1	Continuous support is provided by top and middle management to apply lean practices
LSC2	Perceiving lean implementation as easy and rapid
LSC3	Accepts the changes resulting from Lean adaptation
LSC4	Organizational-wide collaboration in adhering to lean principles
LPA	**Lean Principles Adaptation** [9,33–35]
LPA1	Perceives the organization as maintaining low Work-In-Progress (WIP)
LPA2	Views the organization as adapting Just-In-Time (JIT) practices well
LPA3	Observes frequent continuous improvement activities
LPA4	Experiences strong employee empowerment and involvement in Lean adoption

Table 4
Demographic information.

Demographic variable	Category	Percentage (%)
Gender	Male	57
	Female	43
Age group	25–34	66
	35–44	30
	45 years or older	5
Marital status	Married	41
	Single	59
Highest education level	Graduate Level	100
Job title	Project Manager	27
	Associate Tech Lead	23
	Associate Quality Assurance Lead	14
	Quality Assurance Lead	11
	Associate Project Manager	9
	Tech Lead	9
	Systems Architect	7
Years of experience	3–5 years	64
	>5 years	36
Company size	Medium (100–250 Employees)	55
	Large (>500 Employees)	45

4. Data analysis and discussion

4.1. Assessing the measurement model

The evaluation of the measurement model was conducted using the "SmartPLS 4″ software, to assess the reliability and validity of the reflective constructs and their indicators. The measurement model was evaluated using four main tests to confirm the adequacy of the reflective constructs before proceeding to the structural model analysis. First, internal consistency reliability was assessed using Cronbach's alpha and composite reliability to determine whether the indicators within each construct consistently measured the same underlying concept. Second, convergent validity was examined through outer loadings and the average variance extracted values to verify whether the indicators of each construct were sufficiently correlated and adequately represented their respective latent variable. Third, discriminant validity was assessed using cross-loadings, the Fornell-Larcker criterion, and the Heterotrait-Monotrait Ratio of Correlations to determine whether each construct was empirically distinct from the others. Finally, model fit was evaluated using the standardized root mean square residual, which indicates the degree of discrepancy between the observed correlations and those reproduced by the model. Together, these tests provide a systematic basis for confirming the reliability, validity, and overall adequacy of the measurement model.

4.1.1. Internal consistency reliability

Internal consistency reliability assesses the degree to which results remain consistent across items within a construct. The internal consistency coefficient reflects how closely a respondent's observed score approximates their true score under an ideal measurement condition [53]. To evaluate internal consistency reliability, both Cronbach's alpha (α) and composite reliability (CR) values were examined (Table 5). All Cronbach's alpha (α) values surpassed the acceptable threshold of 0.7 [54], while all composite reliability (CR) values exceeded 0.7, confirming a high level of internal consistency across the dataset [55]. These results indicate that all constructs satisfy the recommended reliability thresholds and exhibit satisfactory shared variance, confirming that the measurement items consistently represent their respective latent constructs.

4.1.2. Convergent validity

Convergent validity refers to the degree to which indicators of a specific construct are positively correlated. It is assessed by examining the outer loadings of the reflective indicators and the Average Variance Extracted (AVE) values [51]. According to [56], indicators with outer loadings below 0.7 should be eliminated. Accordingly, the indicator LSK4 was removed. As presented in Table 6, the final measurement model retained only indicators with outer loadings above 0.7, thereby supporting convergent validity. The retained indicators show strong item reliability, as all outer loadings exceed the recommended threshold, indicating that each item contributes meaningfully to its intended construct.

Furthermore, AVE values exceeding 0.5 indicate that the measurement items adequately represent their respective constructs [57]. As shown in Table 5, all AVE values in the dataset were above 0.5, confirming convergent validity.

Table 5
Reliability and AVE values.

	α	CR	AVE
LSK	0.858	0.913	0.778
SEB	0.937	0.955	0.842
LSC	0.891	0.925	0.755
LPA	0.876	0.915	0.731

Table 6
Outer loading values.

Constructs (C)	Indicators	Outer Loadings
LSK	LSK1	0.872
	LSK2	0.882
	LSK3	0.892
SEB	SEB1	0.942
	SEB2	0.930
	SEB3	0.918
	SEB4	0.879
LSC	LSC1	0.824
	LSC2	0.903
	LSC3	0.878
	LSC4	0.868
LPA	LPA1	0.795
	LPA2	0.857
	LPA3	0.842
	LPA4	0.921

4.1.3. Discriminant validity

Discriminant validity refers to the extent to which a given construct is empirically distinct from other constructs [51]. To evaluate this, three criteria were applied across the three datasets: cross-loadings, the Fornell-Larcker (FL) criterion, and the Heterotrait-Monotrait Ratio of Correlations (HTMT). As shown in Table 7, each indicator demonstrated the highest loading on its intended construct, confirming discriminant validity based on cross-loading analysis [58]. Since each indicator loads highest on its own construct compared to the others, the results provide additional evidence that the constructs are empirically distinguishable.

According to the FL criterion, discriminant validity is established when the square root of a construct's AVE exceeds its correlations with other constructs [59]. This condition was satisfied across the dataset, as indicated in Table 8. The diagonal values are greater than the corresponding inter-construct correlations, further confirming that each construct shares more variance with its own indicators than with other constructs.

Additionally, all HTMT values remained below the threshold of 1.0 (Table 9), further supporting the presence of discriminant validity [51].

4.1.4. Model fit (SRMR index)

The standardized root mean square residual (SRMR) reflects the discrepancy between the observed correlation matrix and the correlations implied by the model and is used to assess overall model fit. The SRMR value of 0.072 for both the saturated and estimated models indicates an acceptable fit, since values below 0.08 are generally considered satisfactory [51].

4.2. Assessing the structural model

Based on the results of the measurement model assessment, it can be

Table 7
Cross loadings.

	LPA	LSC	LSK	SEB
LPA1	0.795	0.568	0.756	0.650
LPA2	0.857	0.625	0.711	0.718
LPA3	0.842	0.611	0.612	0.621
LPA4	0.921	0.722	0.778	0.798
LSC1	0.635	0.824	0.571	0.494
LSC2	0.609	0.903	0.594	0.524
LSC3	0.680	0.878	0.604	0.570
LSC4	0.649	0.868	0.554	0.552
LSK1	0.664	0.523	0.872	0.523
LSK2	0.809	0.569	0.882	0.649
LSK3	0.732	0.675	0.892	0.692
SEB1	0.826	0.662	0.787	0.942
SEB2	0.709	0.593	0.580	0.930
SEB3	0.733	0.562	0.640	0.918
SEB4	0.733	0.439	0.577	0.879

Table 8
Fornell-Larcker criteria.

	LPA	LSC	LSK	SEB
LPA	0.855			
LSC	0.742	0.869		
LSK	0.839	0.669	0.882	
SEB	0.820	0.617	0.709	0.918

Table 9
HTMT values.

	LPA	LSC	LSK	SEB
LPA				
LSC	0.835			
LSK	0.958	0.763		
SEB	0.898	0.671	0.780	

concluded that the model demonstrates acceptable accuracy and reliability across the dataset. Consequently, the evaluation of the structural model was initiated. The hypothesized relationships among the constructs were examined using the structural model in "SmartPLS 4″ software. The structural model results were employed to address the research questions by testing the proposed hypotheses. To assess the model's explanatory power, the coefficient of determination (R^2) value was analyzed. The coefficient of determination (R^2) is the most widely used measure for evaluating the structural model because it indicates the model's explanatory and predictive power with respect to the dependent construct. More specifically, R^2 is calculated as the squared correlation between the actual values and the model-predicted values of a given dependent construct. It therefore reflects the combined effect of all independent latent variables linked to that dependent variable and represents the proportion of variance in the dependent construct explained by those predictor constructs. In this study, R^2 is used to assess how well Lean Specific Knowledge, Short-term Economic Benefits, and Lean Supportive Culture jointly explain the variance in Lean Principle Adaptation. Further, R^2 is interpreted as a measure of the model's in-sample predictive power [51]. The observed R^2 value for the dependent construct was 0.832, indicating adequate variance for the predictor variables [51].

Fig. 2 presents the finalized path model estimated using SmartPLS 4 and visually summarizes both the measurement model and the structural relationships among the study constructs. The values shown on the arrows connecting the independent constructs to the dependent construct represent the standardized path coefficients (β), indicating the strength and direction of the hypothesized relationships. As shown in the figure, all three paths are positive. Lean Specific Knowledge (LSK) records the strongest effect on Lean Principle Adaptation (LPA) with a path coefficient of 0.412, followed by Short-term Economic Benefits (SEB) with 0.388, while Lean Supportive Culture (LSC) shows the smallest, yet still positive, effect with 0.226. These values indicate that all three factors contribute positively to lean principle adaptation, with Lean Specific Knowledge emerging as the most influential predictor in the model. The value shown inside the LPA construct, 0.832, represents the coefficient of determination (R^2), indicating that 83.2% of the variance in Lean Principle Adaptation is explained jointly by LSK, SEB, and LSC. In addition, the values shown between each latent construct and its observed indicators represent the outer loadings of the retained measurement items. These loadings are all above the acceptable threshold, indicating that the indicators are strongly associated with their respective constructs.

4.2.1. *Testing for hypothesized effects of the base model*

To make the hypothesis testing procedure more transparent, the statistical measures used in this section are briefly explained as follows. The hypothesized relationships were assessed using the path coefficient, t-value, p-value, and 95% confidence interval obtained through bootstrapping. The path coefficient indicates the strength and direction of the relationship between an independent construct and the dependent construct. The t-value is used to assess whether the estimated path

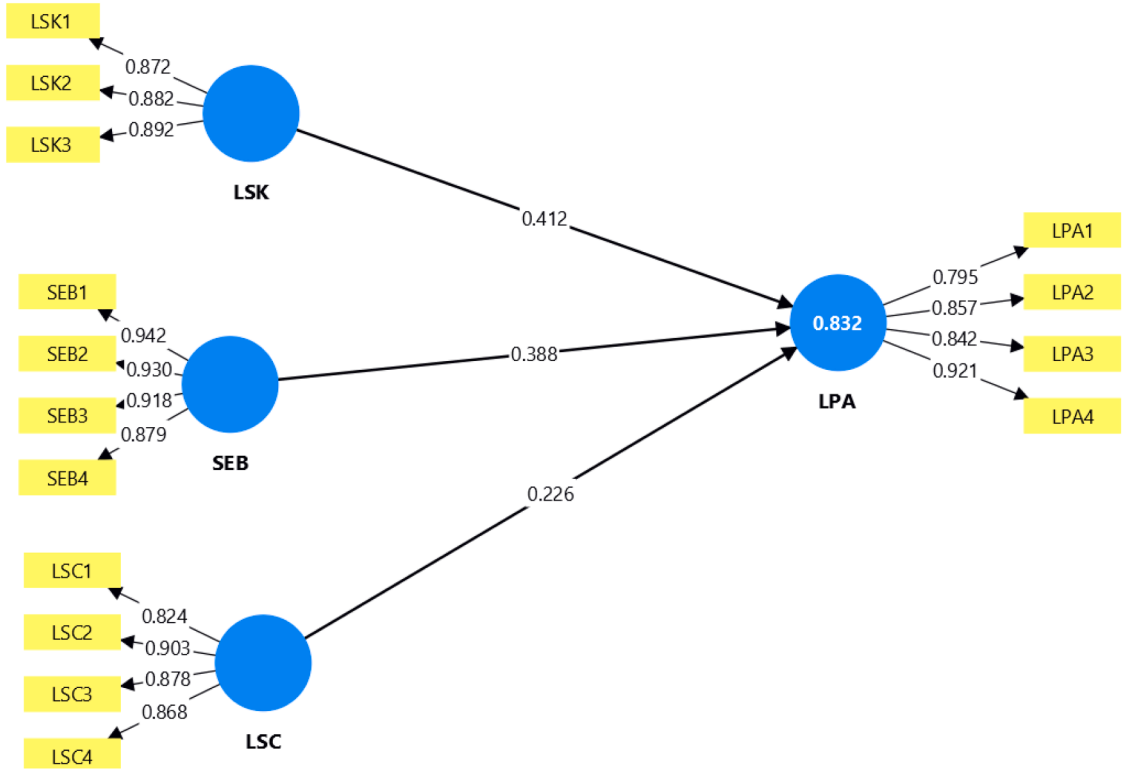


Fig. 2. Path model.

coefficient is sufficiently different from zero. Since the sample size exceeds 30, the t distribution can be approximated by the normal distribution, and a t-value greater than 1.96 indicates statistical significance at the 5% level for a two-tailed test. The p-value represents the probability of observing a result at least as extreme as the estimated one if the null hypothesis of no effect were true. Accordingly, p-values below 0.05 indicate statistically significant relationships. In addition, the 95% confidence interval provides an interval estimate of the path coefficient, and a relationship is considered significant when this interval does not include zero. Together, these measures provide a comprehensive basis for determining whether each hypothesized relationship is statistically supported [51].

These direct effects were assessed using the dataset, and the results are presented in Table 10, which includes Sample mean (M), Standard deviation (STDEV), path coefficients (β), 95% confidence intervals (CI), t-values, and p-values for the three hypothesized direct relationships. Statistical significance was evaluated at the 0.05 level, where p-values <0.05 and t-values exceeding 1.96 indicate significance [51]. Based on these criteria, all three hypotheses were found to be statistically significant within the sample. Further, Table 10 indicates that all VIF values fall below 5 and remain above 0.20 [51], suggesting that predictor collinearity does not pose a concern in the structural model. The f^2 values (Table 10) are in the medium-to-large range [51], indicating that each predictor construct makes a meaningful contribution to the explained variance (R^2) of the target construct in the structural model. Overall, the results in Table 10 confirm that all three predictor constructs exert statistically significant positive effects on lean principle adaptation, with Lean-Specific Knowledge showing the strongest effect among the tested relationships.

5. Discussion

The findings are consistent with prior literature reviewed in Section 2 and summarized in Table 1. In particular, the prominence of Lean-Specific Knowledge aligns with [12], while the significance of cultural and economic factors is consistent with the barriers reported by [9,11,25,26]. Compared with these earlier studies, the present model offers three advantages. First, it integrates three organizational theories within a single framework tailored to software project management. Second, it quantitatively estimates and prioritizes the relative influence of the barriers. Third, it introduces a methodological contribution through the application of PLS-SEM in this research area, particularly within the Sri Lankan software development context.

The hypotheses developed in this study were tested across the dataset to assess the influence of key factors on Lean adoption. All three hypotheses were supported, aligning with and reinforcing findings from previous literature, [9,11,25,26]. Under the positivist framework applied in this research, the statistical confirmation of these positive relationships validates their inverse conditions as significant barriers [54], meaning that the absence or insufficiency of these enabling factors constitutes the critical impediments to Lean implementation.

In addition, the study extends the theoretical understanding of Lean adoption by confirming the relevance and applicability of three major organizational theories: Resource-Based View [41], Contingency Theory [45], and Institutional Theory [48] to the context of software development firms in developing economies. Specifically, the strong support for Lean-Specific Knowledge validates the Resource-Based View by demonstrating that technical know-how is a critical intangible resource in this context, while the significance of Short-term Economic Benefits supports Contingency Theory by suggesting that adoption is contingent upon immediate financial viability. Furthermore, the role of Lean Supportive Culture aligns with Institutional Theory by highlighting the necessity of standardizing management support for authority. By empirically validating these theories within this specific industry and regional context, the study contributes to the expansion of their applicability beyond the traditional manufacturing sector or developed economy settings.

Regarding the prioritization of factors, Lean-Specific Knowledge was identified as the most significant barrier to Lean adoption, followed by Short-term Economic Benefits and Lean Supportive Culture. This hierarchy indicates that the lack of Lean-related knowledge and expertise poses the greatest challenge for software development firms in developing countries, outweighing even financial or cultural constraints. This emphasizes the need for targeted knowledge-building initiatives to support successful Lean implementation, as the primary hurdle appears to be a fundamental gap in knowledge.

6. Conclusions and implications

This study investigated the challenges affecting the adoption of lean principles in software project management within Sri Lanka's software development industry. Guided by two main research objectives, the study identified the main challenge factors, examined their influence on lean principle adaptation, and prioritized their relative significance. Using a structured survey and PLS-SEM analysis, the study found that Lean-Specific Knowledge had the strongest positive effect on Lean Principle Adaptation ($\beta = 0.412$), followed by Short-term Economic Benefits ($\beta = 0.388$) and Lean-Supportive Culture ($\beta = 0.226$), with the model explaining 83.2% of the variance ($R^2 = 0.832$). These findings underline the study's contribution in offering a theory-grounded and empirically validated explanation of lean adoption barriers in a developing-country's software development industry context.

Further, this investigation offers important theoretical and practical implications for Lean adoption within software development firms in developing countries. It is the first to explore the barriers to Lean implementation from a software project management perspective in Sri Lanka. Additionally, it uniquely applies the PLS-SEM methodology, which has not been used in prior local studies. Responding to the need for more quantitative research in this area [11], the study addresses a significant gap in the literature by incorporating three organizational theories: Resource-Based View, Contingency Theory, and Institutional Theory. Each of these theories helps to explain the various challenges that hinder Lean adoption, and expands the theories' applicability to the developing countries' software development industry. On a practical level, the study identifies key obstacles faced by software development firms and offers actionable recommendations for leadership to address these issues effectively. It highlights three major challenges: limited Lean-related knowledge, the lack of immediate and visible financial returns from Lean practices, and organizational cultures that are not aligned with Lean principles. These findings provide a strong foundation

Table 10
Assessing direct effect.

Direct Path	Values							
LSC → LPA	**Mean**	**STDEV**	**β**	**95% CI**	**t**	**p**	**VIF**	**f^2**
(H3)	0.241	0.099	0.226	[0.056, 0.457]	2.285	0.022	1.957	0.156
LSK → LPA	**Mean**	**STDEV**	**β**	**95% CI**	**t**	**p**	**VIF**	**f^2**
(H1)	0.427	0.154	0.412	[0.098, 0.703]	2.669	0.008	2.436	0.414
SEB → LPA	**Mean**	**STDEV**	**β**	**95% CI**	**t**	**p**	**VIF**	**f^2**
(H2)	0.364	0.172	0.388	[0.028, 0.653]	2.254	0.024	2.173	0.412

for developing targeted strategies to support successful Lean implementation in the software industry.

7. Limitations and future work

This study is subject to several limitations. Data collection was conducted cross-sectionally through convenience sampling, and the sample was limited to employees within software development firms. Key external stakeholder groups, such as clients, vendors, and end users, were not considered. Additionally, the study adopted a purely quantitative approach. Future research could address these limitations by incorporating a broader range of stakeholders, undertaking multi-stakeholder analyses, and employing mixed-method approaches that include qualitative insights. Greater emphasis could also be placed on developing actionable recommendations to address the identified challenges. Further, all HTMT values are below the general threshold of 1.0, supporting acceptable discriminant validity in the present model [51], and the HTMT criterion is also unlikely to indicate a lack of discriminant validity when constructs are highly but not perfectly correlated with values close to 1.0 [60]. However, since some HTMT values exceed the more conservative 0.90 guideline used in the literature [61] to further reduce the risk of construct overlap, future work could refine the measurement items and re-validate discriminant validity using stricter cut-offs and an independent sample. The sample size of 44 responses, although exceeding the minimum statistical power requirement (37), remains modest and may limit estimate stability in the multi-construct PLS-SEM model. While the study is positioned as contributing insights relevant to software development firms in developing-country settings, the evidence is derived from a single national context (Sri Lanka) and a limited sample; therefore, the findings may not be interpreted as universal for all developing countries. Future research is encouraged to replicate the model using larger samples across multiple developing-country contexts to strengthen external validity and support broader generalization.

Funding sources

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Compliance with guidelines and regulations

All methods and procedures employed in this study were conducted in full compliance with relevant institutional, national, and international guidelines and regulations, ensuring adherence to the highest standards of research ethics.

Ethical considerations

Approval for this study was obtained from the Department of Industrial Management, University of Kelaniya, Sri Lanka. Before participation, all subjects were provided with detailed information regarding the study's objectives, procedures, and their rights, including the voluntary nature of participation.

Informed consent

Informed consent was obtained from all participants involved in the study. Participants were assured of anonymity and confidentiality, and their right to withdraw from the study at any time without penalty.

CRediT authorship contribution statement

Hasitha Rathnayake: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amila Withanaarachchi:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Amila Thibbotuwawa:** Writing – review & editing, Writing – original draft, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.infsof.2026.108209.

Data availability

The datasets generated or analyzed during the current study are available from the corresponding author, Hasitha Rathnayake (H.R.), upon reasonable request.

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